

平成16年度2月期 筑波大学 大学院博士課程 システム情報工学研究科
コンピュータサイエンス専攻 知能機能システム専攻 構造エネルギー工学専攻
入学試験問題

外国語

えいご
英 語

問題1. 次の英文を読んで、各問いに答えなさい。

Read the following English writing and answer the questions.

The soaring, jagged peaks of the world's second highest mountain range, the Andes, could owe their stature not [①] the power of the Earth but to that of the sea.

Most mountain ranges are created by the collision of two continental plates, such [②] the grinding action of India against Asia that thrust up the Himalayas. But the Andes are perched on a point where an oceanic plate slips down beneath a continental one. Great mountains aren't usually born of such meetings. Heavy, dense oceanic plates tend to slip underneath continental ones, causing major earthquakes but not world-class mountains. If the push of the mid-Atlantic ridge on tectonic plates were the only factor driving up the Andes, one calculation shows that they would be no more than two kilometers high — half their actual height.

Casual links [③] the rise of a large mountain range and climate have often been considered to work in one direction, with significant uplift provoking climate change. Here we propose a mechanism by which Cenozoic climate change could have caused the rise of the Andes. Based on considerations of the force balance in the South American lithosphere, we suggest that the height of, and tectonics in, the Andes are strongly controlled both by shear stresses along the plate interface in the subduction zone and by buoyancy stress contrasts between the trench and highlands, and shear stresses in the subduction zone depend on the amount of subducted sediments. We propose that the dynamics of subduction and mountain-building in this region are controlled by the processes of erosion and sediment deposition, and ultimately climate. In central South America, climate-controlled sediment starvation would then cause high shear stress, focusing the plate boundary stresses that support the high Andes.

(Nature, October 2003 より抜粋)

注) perch: 高い所に置く、Cenozoic: 新生代(約6500万年前から現在に至る地質年代)、lithosphere: 地球表層の(上部)岩石圏、starvation: 窮乏、不足

平成16年度2月期 筑波大学 大学院博士課程 システム情報工学研究科
コンピュータサイエンス専攻 知能機能システム専攻 構造エネルギー工学専攻
入学試験問題

外国語

えい	ご
英	語

問(1). 上記の英文に、表題をつけなさい。

Question (1). Put a title to the above English writing, in Japanese or English.

問(2). 括弧内に最も適切な前置詞を入れて、理解できる文章を完成しなさい。

Question (2). Answer the most appropriate prepositions in brackets to make up comprehensible sentence.

[①]

[②]

[③]

問(3). アンデス山脈の高い高度を支配している要素を、日本語か英語で80語以内で説明しなさい。

Question (3). Give an explanation for some factors having dominated the high rise of the Andes, in Japanese or English (in less than 80 Japanese characters or English ones).

外

平成16年度2月期 筑波大学 大学院博士課程 システム情報工学研究科
コンピュータサイエンス専攻 知能機能システム専攻 構造エネルギー工学専攻
入学試験問題

外国語

えい 英	ご 語
---------	--------

問題2. 次の専門語を、英語で簡単に説明しなさい.

Question 2. Make a brief explanation for the following technical terms in English

問(1) 多面体 (polyhedron)

問(2) 正三角形 (equilateral triangle)

問(3) 硬度 (hardness)

問(4) 空気 (air)

問(5) ガラス (glass)

問(6) 熱膨張 (thermal expansion)